

Ex.No: 9

Demonstration of Intrusion Detection System (IDS) Using Snort

AIM:

ALGORITHM:

CODE:

REM 1. Navigate to Snort directory

cd C:\Snort

REM 2. Verify Snort installation

bin\snort.exe -V

REM 3. Display available network interfaces

bin\snort.exe -W

REM 4. Basic Sniffer Mode

REM Replace 5 with your actual interface number

bin\snort.exe -v -i 5

REM Press Ctrl+C to stop Snort before continuing

REM 5. Display packet header and payload

```
bin\snort.exe -vd -i 5
```

REM Press Ctrl+C to stop Snort before continuing

REM 6. Display packet in hexadecimal format

```
bin\snort.exe -X -i 5
```

REM Press Ctrl+C to stop Snort before continuing

REM 7. Create packet log directory

```
mkdir log
```

REM 8. Run packet logging

```
bin\snort.exe -dev -l log -i 5
```

REM Generate some traffic, then press Ctrl+C

REM 9. Check packet log

```
dir log
```

REM 10. Create custom ICMP rule

```
cd C:\Snort\rules
```

REM Open Notepad

```
notepad local.rules
```

REM Enter the following rule in Notepad:

```
REM alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

REM Save as:

```
REM local.rules
```

REM 11. Start Snort IDS

```
cd C:\Snort
```

```
bin\snort.exe -A console -q -i 5 -R rules\local.rules
```

REM Generate ICMP traffic:

```
ping 8.8.8.8
```

```
REM [**] [1:1000001:1] ICMP Ping Detected [**]
```

```
Copyright (C) 1998-2018 Sourcefire, Inc.  
Using libpcap version 1.10.1 (with TRACKET_V3)  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.2.11
```

Snort successfully initialized (Version 2.9.20)

```
C:\Snort\bin>snort.exe -W
```

```
o--'-  
'---' )~-'--'  
      / \   \|/\|  
    --(X)-
```

```
-*> Snort! <*~  
Version 2.9.20-WING4  
By Martin Roesch & The Snort Team
```

Available network interfaces:

1. [\Device\NPF_{A2C8F7D1-5E68-4F2A-8C3D-7E9A1D2B4C6F}] (Realtek PCIe GbE Family Controller)
2. [\Device\NPF_{6E7F8A9B-1C2D-4E5F-8A9B-0C1D2E3F4A5B}] (Loopback Pseudo-Interface 1)

Enabling promiscuous mode on interface 1 (Realtek PCIe GbE Family Controller)...

```
C:\Snort\bin>
```

```
C:\Snort\bin>snort.exe -i 1 -v
```

```
08/12-14:32:17.123456  [**] [1:1000001:1] ICMP Ping [**]  
[Priority: 0]  
{ICMP} 192.168.1.10 -> 192.168.1.1  
Type: 8 Code: 0 ID: 0 Seq: 1
```

```
Incoming packet detected ---
```

Index	Physical Address	IP Address	Device Name	Description
1	00:00:00:00:00:00	disabled	\Device\NPF_{9096A3A2-0F91-4748-A9D6-B627897D533F}	Network Monitor)
2	00:00:00:00:00:00	disabled	\Device\NPF_{48973314-849C-4F99-B527-A23E2D54F25E}	v6)
3	00:00:00:00:00:00	disabled	\Device\NPF_{83BBA936-4B15-4566-8EB5-F08CEE7856D2}	)
4	00:15:5D:EB:E0:65	172.20.224.1	\Device\NPF_{E388DC4A-ACEF-400D-874A-37874A0D4F1A}	Ethernet Adapter
5	88:AE:DD:15:EC:36	172.16.53.200	\Device\NPF_{8FEAF4C3-A870-4EE3-ADE8-AADFC638F850}	Family Controller
6	00:00:00:00:00:00	0000:0000:0000:0000:0000:0000	\Device\NPF_Loopback	Adapter

RESULT: